

AI & Knowledge Management for Researchers

What generative AI actually is, which tools matter, and how to make them answer faithfully from your documents.

BEFORE WE START

Three questions for the room

1

Did you use an AI tool this week? For what?

2

What is one thing you don't trust about AI?

3

What is one task in your research you would gladly hand to a capable assistant?

Collect answers on a whiteboard; revisit at the end.

TODAY • 2 HOURS

Where we're going

Part
1

AI 101

What this technology actually is

~30 min

Part
2

The main kinds of tools

Tools that answer, tools that act, and tools that remember

~25 min

Part
3

Hands-on

Make AI answer from YOUR documents

~50 min

Part
4

Caveats and takeaways

What to carry out of the room

~15 min

| You will leave with one working AI notebook that answers from your own documents, built by you and checked by you.

PART 1 • AI 101

Generative AI is a new kind of tool, and like every tool, it has a **shape.**

We have always built tools that **extend us**

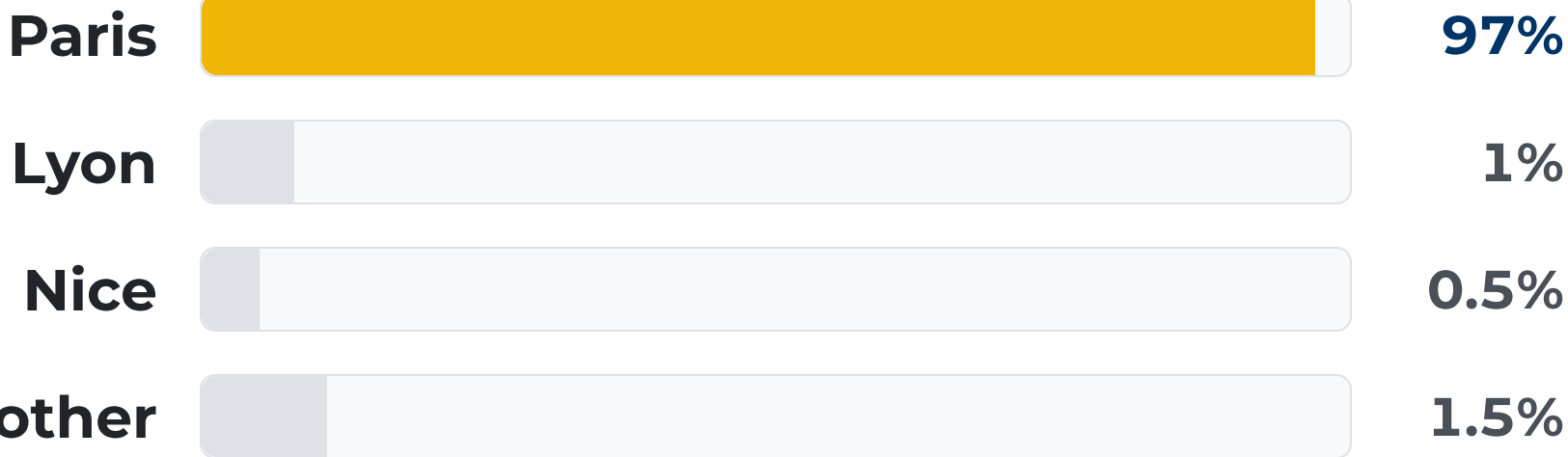


AI sits on a long continuum. What is new is that it works in natural language, which is your medium as a researcher.

Under the hood, the model predicts the **next word**, and it does this astonishingly well

“The capital of France is _____”

NEXT-TOKEN PROBABILITIES · ILLUSTRATIVE



Token

The unit of text the model reads and writes, roughly a fragment of a word.

Model

The trained statistical system that produces these probabilities.

Context window

How much text it can “see” at once when it predicts.

There is no database lookup behind the answer and no guarantee of understanding. The model produces a well-calibrated guess from patterns in its training data.

When the patterns are thin, it guesses, and it does so **confidently**

● ● ● AI ANSWER

“Cite a study on this effect.”

Certainly. A frequently cited source is:

Hartman, L. & Okonkwo, R. (2019). “Longitudinal effects of grounding on recall.” **Journal of Cognitive Methods**, 14(3), 221–248.

FABRICATED

**IT READS AS FLUENT AND PLAUSIBLE, YET
THE SOURCE DOES NOT EXIST**

FROM AI 101

The risk is not that AI hallucinates. It is that AI gets deployed **without grounding.**

Grounding

Requiring the model to answer from sources you supply, with citations you can check.

People shaped raw prediction into an **assistant**

1

PRETRAINING

Read most of the internet

It learns the patterns of language from a vast sweep of text.



2

INSTRUCTION TUNING

Learn to follow requests

It is shaped to take an instruction and actually answer it.



3

HUMAN FEEDBACK

Learn what people prefer

People rate its answers, and it is tuned toward the ones they like.

Helpfulness and tone are **trained into the model** by its makers. The assistant is a designed artifact, and its designers' choices are built into how it responds.

It works best where the **data was thickest**

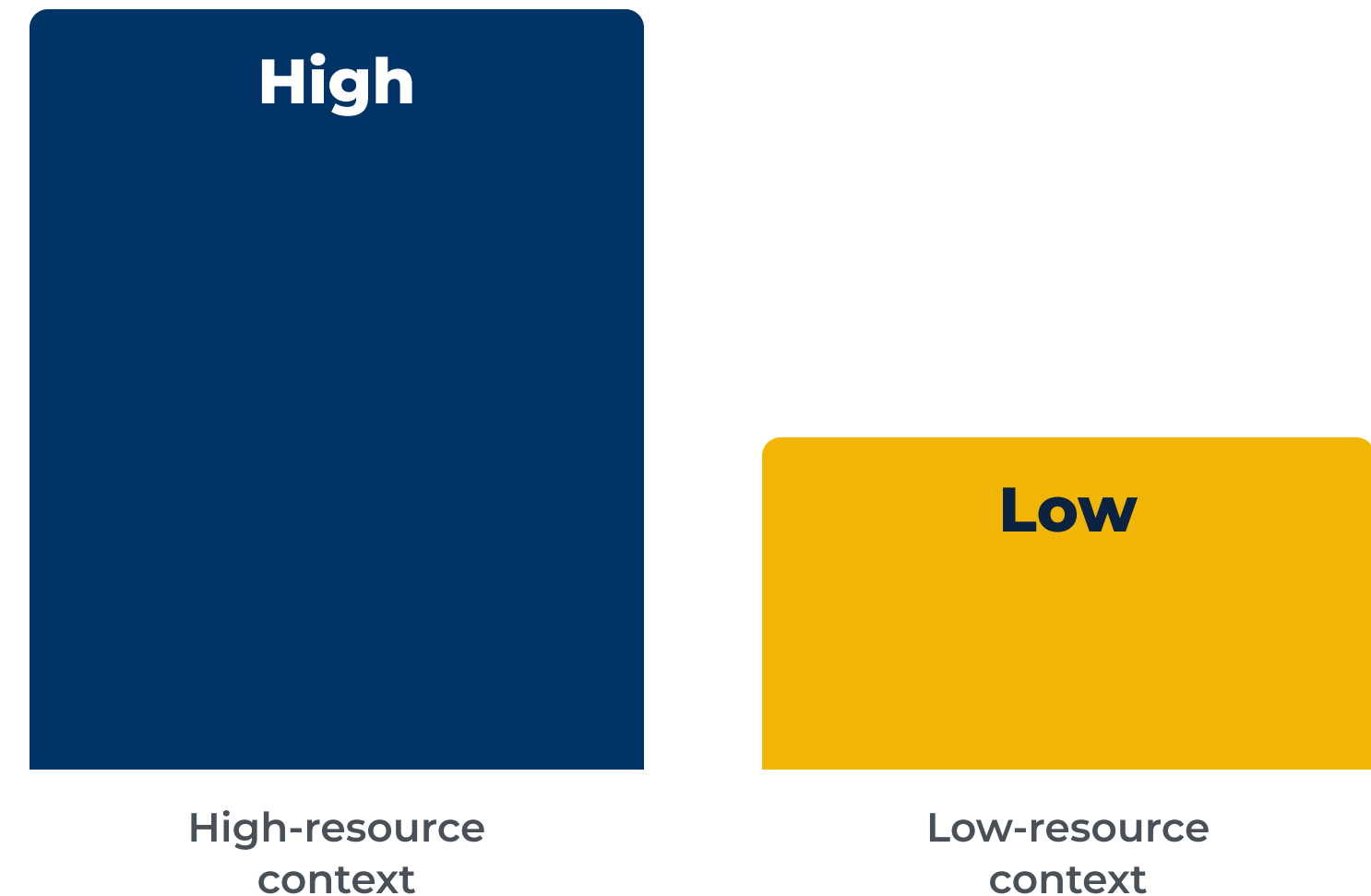
Performance varies sharply by language and domain. A high benchmark score reflects average conditions, so it can overstate how the model behaves in your field. For **underrepresented languages and communities**, behavior tends to be weaker and less safe.

THE RESEARCHER ANGLE

Your niche subfield is a “**low-resource language**” too, so you should verify its answers more carefully.

Reliability by context

ILLUSTRATIVE



“Seems fine” is **not an evaluation**

Level 1

Models Evaluation

Does the AI system perform as intended?

Level 2

Product Evaluation

Does the overall product engage and retain users?

Level 3

User Evaluation

Does the product change users' thoughts, feelings, knowledge and behaviour towards the development outcome?

Level 4

Impact Evaluation

Do users with access to the product improve development outcomes?

Today's hands-on work lives at **Level 1**: you will test whether an AI system performs as intended on your own documents.

Read the data terms **before** you upload anything sensitive

CONSUMER FREE TIER

May train on your chats

Free tiers can use your conversations to train future models. Check the settings and opt out if you can.

ENTERPRISE / WORKSPACE TIER

Generally will not

Paid organizational tiers typically do not train on your data, but you should confirm what the contract says.

IRB-SENSITIVE

Approval first

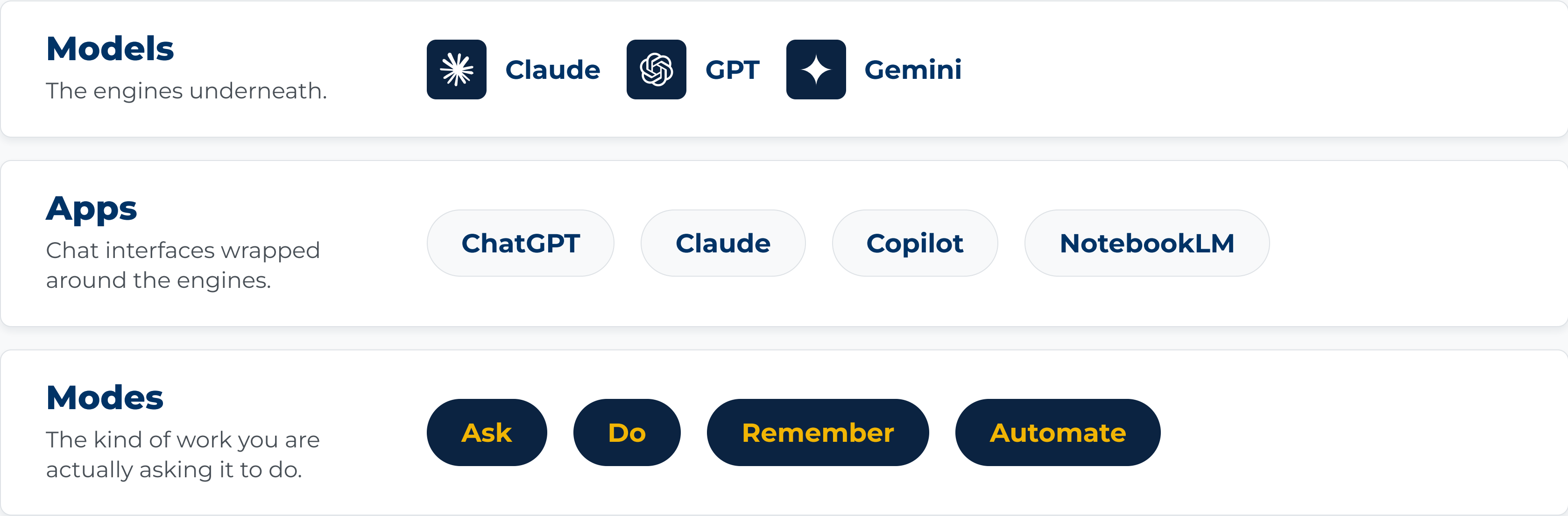
IRB-covered, identifiable, or unpublished sensitive data needs institutional approval before upload.

TODAY Bring only documents you'd be comfortable **emailing to a colleague**.

PART 2 • THE TOOL LANDSCAPE

The tools are converging on four capabilities: ask, do, remember, and automate.

The confusion is real, and it has a clear structure



Track the capabilities rather than the individual products. **Products change from month to month, but the underlying capabilities stay stable.**

NotebookLM: a librarian that only cites **your shelf**



NotebookLM

WHAT IT DOES

Upload up to **50 sources** on the free tier: PDFs, Docs, links, and audio. When you ask a question, **every answer cites the passages** it drew from, and you can click each one to verify it.

STUDIO EXTRAS

Audio Overviews (podcast-style discussion of your sources), mind maps, and flashcards.

WHY RESEARCHERS CARE

It is a **closed system**: it answers from the documents you give it, not from the open internet. This is what grounding looks like as a finished product.

● YOUR NOTEBOOK · 4 SOURCES

YOU ASK

What did the studies conclude about the intervention's effect?

GROUNDING ANSWER

Both sources report a positive effect, with the larger study noting the result held only in the follow-up cohort.

[1] Source A · p.4

[2] Source B · p.11

Every claim links to a passage you can check.

Every major assistant has an **Ask** mode: you converse, it answers



Claude

Strong with long documents and careful reasoning



ChatGPT

The most widely used; broadest feature set



Gemini

Lives inside Google Workspace; reads very long context



Microsoft Copilot

Built into Word, Excel, Outlook, and Teams

Each one rests on the same idea: a chat window placed over a large language model. Your choice usually follows where your documents already live.

A Do mode acts on the work itself: it **does the work**



Claude Cowork

Point it at a folder; it reads, organizes, drafts, and returns a finished deliverable.

Built for non-programmers



Claude Code / Codex

Terminal agents that run multi-step tasks end-to-end.

Built for technical workflows, and increasingly used beyond code

"What should I write?" → "Draft it, I'll review."

THE TRADE

The more autonomy you grant, the more **carefully you must verify** the output. You review what it produces, every time.

Ask when you're **thinking**; Do when you're **producing**

ASK you're thinking

- Exploring a topic
- Checking your understanding
- Low stakes per answer

DO you're producing

- Repetitive, multi-file work
- A defined deliverable
- You can verify the result

BOTTOM LINE

Delegate a task only when you can **verify the result** yourself.

"Remember" modes hold your context so you need not **repeat it** each time



Claude Projects

Upload files + instructions once; every chat in the project knows them.



ChatGPT Projects & Custom GPTs

Grouped chats with files; shareable custom assistants.



Gemini Gems

Custom assistants that sync with Google Drive, so their files stay current.



Microsoft Copilot

Works inside your existing M365 permissions.

| At this point an AI tool begins to function as a knowledge-management system.

A fourth capability is emerging: AI that **automates** work on your behalf

Beyond asking, doing, and remembering, the newest systems learn enough about your work to carry out **recurring tasks on their own**, on a schedule or in response to events, under supervision rules that you define. You set the standing instructions once; the system runs and reports back.

FOR A RESEARCHER

A weekly scan of new papers in your subfield, summarized and delivered every Monday morning.

FOR A TEAM

Meeting recordings transcribed, summarized, and filed to the right project folder without anyone asking.

FOR A STUDY

A data-quality check that runs after each survey wave and flags exceptions for human review.

OUTSIDE TODAY'S SCOPE

We will not cover automation today. The principle you will practice applies here with greater force: the more a system acts **for** you, the more deliberately you must verify its work.

Our recommendation: choose by task, then anchor on one primary tool

Analyzing long documents and writing with care	→	 Claude	Strong sustained reasoning over book-length material. Projects hold your sources and standing instructions.
Question-answering over a fixed set of sources	→	 NotebookLM	A closed system by design. Every answer carries citations that link back to a passage you can check.
Your documents already live in Google Workspace	→	 Gemini	Gems stay synced with Drive, so the assistant reads the current version of every file.
Your institution runs on Microsoft 365	→	 Copilot	Works inside Word, Excel, Outlook, and Teams, governed by the permissions your organization already set.
You want the largest community and guides	→	 ChatGPT	The most widely used assistant, with the deepest pool of tutorials, templates, and shared custom assistants.

Commit to one primary tool for at least a quarter and learn its project features well. When you do re-evaluate, test it on your own tasks with your own questions rather than relying on launch-day announcements.

PART 3 · HANDS-ON

You believe what you **verify**, so let us **verify** it together.

Non-sensitive documents only.
Nothing you couldn't email a colleague.

Work in pairs. Two people notice more than one.

Write down every failure. Each failure is useful data.

ACTIVITY 1 • 10 MIN

Prompt makeover

See how much of an answer's quality comes from the quality of your request.

- 1 Open an AI tool **you already know** (ChatGPT, Claude, Gemini, or Copilot)
- 2 Write a quick, minimal version of a real request from your work, and run it
- 3 Rewrite the request with four elements: who you are, the context that matters, the format you want, and one example
- 4 Run it again and compare the two outputs side by side

DEBRIEF

What changed most: the accuracy, the tone, or the usefulness?

Build your grounded Q&A

Build a working AI interface that answers only from your own documents.

- 1 Gather **as many documents as are useful**: papers, notes, reports, anything you know well
- 2 Create a NotebookLM notebook (or a Project in your familiar tool) and upload them
- 3 Ask **five real questions** you'd ask a colleague who had read them
- 4 Click every citation and confirm that the quoted passage genuinely supports the answer

VALIDATION CHECKLIST

- Is the citation real?
- Does the passage say what the answer claims?
- What did it miss?

Red-teaming your own bot

Try, deliberately and systematically, to make the grounded assistant you just built give a wrong answer.

- 1** Ask a question your documents **cannot** answer. Does the assistant admit the gap, or does it improvise?
- 2** Ask a question with a false premise built in, such as a finding your documents contradict. Does it push back?
- 3** Ask it to overstate: request a stronger conclusion than the evidence supports, and watch how far it goes
- 4** For every answer, open the citations and check whether the quoted passage actually says what the answer claims

DEBRIEF

Where did your assistant break first, and how convincing did it sound while breaking?

KEEP IN MIND

Grounding narrows the room for error; it does not **remove it**. A system you have red-teamed yourself is a system you can use with judgment.

SHARE-OUT

What **surprised** you?

1

One thing that went better than you expected

2

One thing that fell short of what you hoped for

3

If you could design the ideal AI tool for your own work, what would it do?

BACK TO THE WHITEBOARD

Return to the doubts we collected at the start of the session. Did anyone's view **shift**, in either direction?

Three caveats to carry out of this room

1

Tooling

Products change often, but the underlying capabilities (**ask, do, remember**) stay stable. Choose by where your documents live, and re-evaluate quarterly at most.

2

Quality

“Seems fine” is not evidence.

Keep the test questions you red-teamed with today, and run them again whenever a tool ships a new model.

3

Agency

AI produces the draft, and **you make the decisions.**

Once you stop verifying its output, the work is no longer research.

A last idea to take home: **System 3**

Wharton researchers Steven Shaw and Gideon Nave argue that AI now forms a third system of thinking alongside Kahneman's fast intuition and slow deliberation. **Offloading** a task to AI, the way you hand arithmetic to a calculator, keeps your reasoning in control. **Cognitive surrender** is different: the AI makes the decision and you adopt it as your own without noticing the transfer. In their experiments, people's confidence rose whether the AI was right or wrong.

- agenticworkflowsclub.org/ai-101, today's Part 1 in full
- "Thinking, Fast, Slow, and Artificially", Wharton Executive Education, May 2026
- Zezhen (Michael) Wu, The Agency Fund

Offload the work whenever it helps. Never surrender the judgment: decide, deliberately, **which decisions remain yours.**